Statistical Quality Control
Seminar Paper on Profile Monitoring

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June 22, 2025

Contents

1 Introduction **3**
1.1 Profile Monitoring . . . 3
1.2 Overview of Existing Research . . . 4

2 Theory on Profile Monitoring **5**
2.1 Types of Profiles . . . 5
2.1.1 Linear Profiles . . . 5
2.1.2 Non-linear Profiles . . . 6
2.1.3 Multivariate Profiles . . . 6
2.1.4 Geometric Profiles . . . 6
2.1.5 Binary Response Profiles . . . 7
2.2 Phases of Linear Profile Monitoring . . . 7
2.2.1 Phase I: Linear Profile Establishment . . . 7
2.2.2 Phase II: Online Linear Profile Monitoring . . . 8
2.3 Control Charting Techniques for Linear Profiles . . . 9
2.3.1 Parameter-Based Control Charts . . . 9
2.3.2 Other Approaches . . . 10
2.4 Performance Measures for Linear Profiles . . . 11

3 Practical (R simulations) on Linear Profile Monitoring **12**
3.1 Description of the Dataset . . . 12
3.2 Information about the Selected R Packages . . . 13
3.3 Application of Profile Monitoring and Steps . . . 13
3.3.1 Phase I: Profile Establishment . . . 13
3.3.2 Phase II: Online Profile Monitoring . . . 14
3.4 Explanation of Results . . . 15

4 Conclusion **16**

5 Bibliography **17**

1 Introduction

Statistical Quality Control (SQC) has long been a foundation for manufacturing and service industries, providing tools to monitor, control, and improve process performance. A key definition of quality, as stated by Montgomery (2009), is that quality is inversely proportional to variability. This implies that the main concern for practitioners should be the reduction of variability in key quality characteristics. Statistical Process Control (SPC) is a specific branch within SQC, this is a powerful collection of problem-solving tools useful in achieving process stability and improving capability through the reduction of variability, according to Montgomery (2020). SPC's primary focus is on monitoring and controlling a process over time, with control charts as its main technique.

Traditionally, SPC methodologies, such as Shewhart control charts, have focused on monitoring single quality characteristics, like the mean or variance. These charts, first introduced by Walter A. Shewhart in 1924 , were helpful in distinguishing random, common causes of variation from assignable or special causes of variation. Their assumption is that the plotted statistic is computed from independently and identically distributed random variables.

1.1 Profile Monitoring

This is where Profile Monitoring comes into play. As defined by Huwang & Liao (2023), Profile Monitoring is a statistical process control method used to assess the stability of the relationship between a dependent variable and one or more independent variables over time. In this context, a "profile" refers to a functional relationship, typically denoted as y=f(x;β), where y is a dependent variable, x represents one or more independent variables, and β is a vector of unknown parameters that define the specific form of the relationship.

Profile monitoring is a specialized SPC technique dedicated to monitoring the stability of these functional relationships over time. The necessity of profile monitoring arises from situations where changes in the underlying functional relationship, rather than just shifts in individual variable means or variances, are indicative of process degradation or the presence of assignable causes.

Examples of processes where profiles are relevant: Calibration Applications: Calibration curves are a clear and fundamental example of linear profile monitoring. This is because they inherently describe a linear relationship between dependent and independent values. Deviations from this expected straight line are critical for instrument performance and are widely understood.

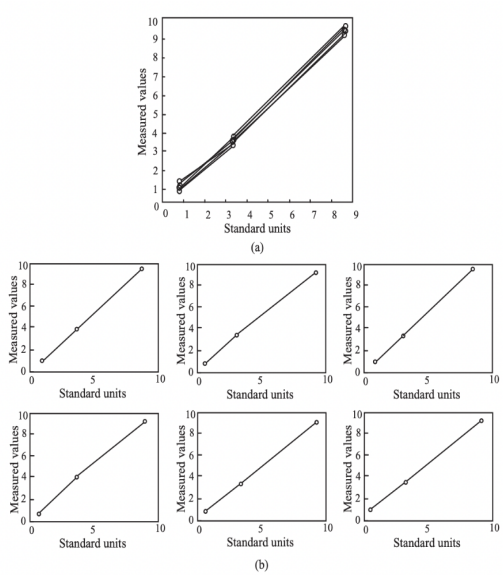

Figure 1: Calibration Curves (adapted from Saghaei & Noorossana 2011.)

The figure is divided into two parts:

- (a) All samples in one figure: This top graph shows multiple calibration profiles on a single plot. Each line represents a different sample, plotting the measured values against the standard units. This gives an visual impression of the consistency of the calibration process across different samples.
- (b) Samples in separate figures: This bottom section presents individual calibration profiles in separate smaller graphs. Each small plot shows the measured values versus standard units for a single sample. This allows for a more detailed inspection of each individual profile, making it easier to identify specific deviations or patterns within a single calibration instance.

The relationship being monitored in calibration is often assumed to be linear, where measured values (y_{ij}) are related to standard values (x_i) through a functional relationship, including error terms.

1.2 Overview of Existing Research

While classical control charts have been foundational to SQC for decades, the field of profile monitoring is a relatively newer development, gaining significant attention over the past 20 years. Profile monitoring is a newer, more relevant way to manage today's complicated processes. Unlike older control charts that look at individual measurements separately, this method takes a different approach. It focuses on keeping the whole process stable, instead of just assuming that if individual parts are stable, that would guarantees overall process quality.

Early work, such as that by Kang and Albin (2000), significantly contributed to establishing methodologies for monitoring simple linear profiles. Initially, the focus was on simpler linear relationships. But over time, this expanded to include more complex curved relationships (non-linear profiles) and situations where multiple factors are monitored at once (multivariate profiles). This ongoing work has also led to different ways of using control charts, such as looking at parameter-based approach or even residual-based ones, each designed for different kinds of profiles and monitoring goals.

While profile monitoring is a powerful , it is not applicable to all functional relationships. It is important to distinguish between true profiles and other types of data, such as time series which does not fit the definition. According to Woodall (2007), functional relationships cannot be considered profiles when time series data are collected on individuals while the time index remains the same. For instance, customer telephone usage patterns over time, though appearing as curves, are typically multiple time series observations of different customers, not successive instances of a single, repeatable function whose parameters are being monitored. In this senario the data represents a unique time series for each individual entity , rather than a series of observations from the same underlying functional form measured repeatedly, then traditional time series analysis or other methods may be more appropriate than profile monitoring.

2 Theory on Profile Monitoring

The types of profiles relates to what kind of model is used to characterize the quality relationship, while the phases of profile monitoring describe how and when that chosen model is applied and utilized over time. The type of profile also directly selected which control charting technique is appropriate and how that technique is mathematically formulated.

2.1 Types of Profiles

The concept in profile monitoring revolves around the portrayal of a process's quality through a functional relationship. Here is a breakdown of the types of profiles:

- **Linear and non linear profiles:** Both have a single dependent variable that is related to one or more independent variables. The key difference is the shape of that relationship: linear profiles gives a straight line , while non-linear profiles result in a curve.
- **Multivariate Profiles:** Has multiple dependent variables and each of these variables is related to one or more independent variables. The primary focus is on monitoring these multiple variables simultaneously, is especially for their correlation.
- **Binary response profiles:** The probability of a specific outcome as a function of one or more independent variables. This implies the use of models like logistic regression.
- **Geometric Profiles:** Describes the shape or form of a product as the quality characteristic. Involves measurements in two or three dimensions, such as roundness, flatness or straightness

2.1.1 Linear Profiles

In the scenario of linear profiles, the quality characteristic is assumed to follow a linear relationship between a dependent variable and one or more independent variables. The general form of a simple linear profile for the i-th observation within a given profile is presented as:

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

In this model:

- y_i represents the i-th observed dependent value.
- x_i denotes the i-th value of the independent variable (often a controlled or measured input).
- β_0 is the intercept parameter, representing the expected value of the dependent variable when the independent variable x_i is zero.
- β_1 is the slope parameter, indicating the expected change in the dependent variable for a one-unit increase in the independent variable.
- ϵ_i represents the random error term, essentially the "noise" of the relationship

For the proper application of regression analysis in profile monitoring, several key assumptions about the error term ϵ_i are typically made:

- Normality: The error terms are often assumed to be following a normal distribution with a mean of zero and a constant variance, $\epsilon_i \sim N(0, \sigma^2)$.
- Independence: The errors are often assumed to be independent.
- Homoscedasticity: The variance of the error term (σ^2) is assumed to be the same across all profiles for a given set of parameters.

Linear profiles are widely applicable in various industrial and service settings. For example, in a manufacturing process, the hardness of a material (y) might linearly increase with the curing temperature (x) and another instance is in a chemical experiment, the yield of a reaction (y) could be linearly related to the concentration of a catalyst (x). Kang and Albin (2000) have notably focused on monitoring linear profiles. These methods often fit separate linear regression models and monitor their coefficients to determine outlying profiles, simplifying the monitoring scheme.

2.1.2 Non-linear Profiles

When the relationship between the dependent variable and the independent variable(s) deviates from a straight line, non-linear profiles come into play. These profiles are characterized by a non-linear functional form, often reflecting the complexities of underlying physical, chemical, or biological occurrences. The general representation for a non-linear profile for the jth measurement of the ith profile is given by:

$$y_{ij} = f(x_{ij}, \theta_i) + \epsilon_{ij}$$

Here,

- $f(\cdot)$ is some nonlinear function,
- x_{ij} is a regressor variable
- θ_i is a vector of parameters for each profile, the parameters here do not always have simple interpretations like intercept and slope, but they define the specific shape and position of the curve.
- ϵ_{ij} represents the random error term, typically assumed to follow the same properties of normality, independence and Homoscedasticity.

Examples of non-linear functions include: exponential growth curves which is commonly seen in biological processes or chemical reactions where a logistic curves which frequently used model for dose-response studies. Monitoring non-linear profiles presents challenges, particularly with large, potentially correlated input data. Previous work often assumed independent measurements within profiles, which is unrealistic , and current approaches may rely on expert judgment, potentially using less of the data.

2.1.3 Multivariate Profiles

In the case of multivariate profiles, there are multiple dependent variables are simultaneously related to one or more independent variables. Whereas previously we had a single dependent variable that is related to one or more independent variables. This requires complex statistical techniques that takes into account the correlations and interdependencies among the multiple dependent variables.

Parker et al. (2001) provide a calibration example, in wind tunnel experiments at NASA Langley Research Center, the force balance involves simultaneously measuring three orthogonal force components (representing dependent variables) and three orthogonal torque components (representing independent variables). The variables involved has a correlation structure between the dependent variables, making this a case of a multivariate multiple linear profile. It is important to account for this correlation, as ignoring this can lead to misleading results.

2.1.4 Geometric Profiles

Beyond one-dimensional functional relationships, profile monitoring extends to scenarios where the quality characteristic is defined by the geometric shape of a product. Based on Colosimo, Pacella, & Semeraro, Q. (2006) the quality of mechanical components is related to both dimensional and geometric specifications. These geometric specifications is a combinations of properties such as straightness, roundness, cylindricity, and flatness. In such cases, the objective is to detect any change from a predefined baseline shape of the product. Normally, in single variable or a multivariate random variables it states that a process is in control if its key features, like its average or spread, stay consistent over time. But with geometric profiles, the real focus is on whether the entire shape of something stays stable over time.

2.1.5 Binary Response Profiles

Profile monitoring applications are not restricted to regression models with continuous dependent variables. According to Saghaei & Noorossana (2011), in many practical applications, the quality of a process or product can be well characterized by a binary dependent variable, taking the form of a conforming or nonconforming product. In such cases, the profile of interest is defined by the probability of a specific outcome - success or failure as a function of one or more independent variables.

To develop control charting techniques suitable for monitoring profiles with binary responses, logistic regression models are typically used. For example, the compressive strength of an alloy fastener used in aircraft manufacturing. Here, the dependent variable is binary (fail or stand the load) when tested at various load strengths (independent variable), forming a profile of failure probability.

For this paper we will be diving deep into the concept of linear profiling as it is the most common and extensively studied type of profile monitoring.

2.2 Phases of Linear Profile Monitoring

Profile monitoring, similar to other SPCs, is implemented in two phases: Phase I (Profile Establishment) and Phase II (Online Profile Monitoring). This structured approach ensures that a stable base is first established, against which future process behaviors can be compared.

2.2.1 Phase I: Linear Profile Establishment

The main goal of Phase I in profile monitoring is to use historical data to establish a stable baseline for the process. This involves identifying and removing any assignable causes of variation that may have been present in the historical data, allowing for an accurate picture of the in-control profile and its inherent variability.

Parameter Estimation: For linear profiles, such as $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$, the parameters β_0 (intercept) and β_1 (slope) for each profile are typically estimated using Ordinary Least Squares (OLS) regression. They are needed because the true, underlying parameters of the linear relationship are unknown in practice, so they must be estimated from observed data for each profileGiven a set of n observations for a single profile, $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$, the OLS estimators for β_0 and β_1 are found by minimizing the sum of squared errors, $\sum_{i=1}^{n}(y_i - (\beta_0 + \beta_1 x_i))^2$. The resulting estimators, denoted as $\hat{\beta}_0$ and $\hat{\beta}_1$, are:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$
$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

where $\bar{x}$ and $\bar{y}$ are the sample means of the independent and dependent variables, respectively..OLS estimators are unbiased, which means that, on average, the estimated values match the actual population parameters. They also have the smallest variance among all linear unbiased estimators, making them the Best Linear Unbiased Estimators (BLUE).According to Moheghi, Noorossana, & Ahmadi (2020), However, in practice, outliers could be present in the data, which would impact the performance of these estimators alot. To overcome this, robust parameter estimation methods, such as M-estimators, are used in Phase I analysis for linear profiles, leading to better estimates and improved Average Run Length performance compaired to traditional Least Squares estimators.

Variance-Covariance Matrix: This matrix is essential because it defines the expected "wobble" of the profile's parameters when the process is stable, allowing control charts to distinguish between normal variation and actual shifts.For the vector of parameter estimates $\mathbf{Z}_j = (\hat{\beta}_0 j, \hat{\beta}_1 j)^T$ for sample j, its variability and correlation are captured by the variance-covariance matrix of the parameter estimates, denoted as $\mathbf{S}$.

When the true error variance σ^2 is known, this matrix is given by:

$$\mathbf{S} = \begin{pmatrix} S_0^2 & S_{01}^2 \\ S_{01}^2 & S_1^2 \end{pmatrix}$$

where:

$$S_0^2 = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right) \text{ is the variance of the intercept estimator } \hat{\beta}_0 j.$$

$$S_1^2 = \sigma^2 \frac{1}{S_{xx}} \text{ is the variance of the slope estimator } \hat{\beta}_1 j.$$

$$S_{01}^2 = -\sigma^2 \frac{\bar{x}}{S_{xx}} \text{ is the covariance between } \hat{\beta}_0 j \text{ and } \hat{\beta}_1 j.$$

The error variance σ^2 is typically unknown and must be estimated from the data. In Phase I, when multiple samples are available, an overall MSE is often computed as the average of individual sample MSE_j values.

Retrospective Analysis: Control charts are applied to the Phase I data to analyze the process. The purpose is to detect any assignable causes that might have been present during the data collection period. One approach involves using a Hotelling's T^2 chart, which is a multivariate control chart designed to monitor the vector of estimated regression coefficients simultaneously. For Phase I, a modified T^2 statistic can be used to compare each historical profile's parameters to the overall estimated in-control parameters, helping to identify outliers in the historical data.. A more detailed explanation of the Hotelling's T^2 chart and its application in profile monitoring can be found in Section "Parameter-Based Control Charts."

This iterative process of detection, investigation, and potential removal leads to the re-estimation of the profile parameters (β_0, β_1) and recalculation of the control limits, ensuring that the final baseline is based on truly in-control data.

While the fundamental steps of parameter estimation, variance-covariance matrix calculation, and retrospective analysis form the core mechanism of Phase I in profile monitoring, there are a few other additional steps to complete the phase I analysis. These can include initial data visualization and exploratory data analysis to identify obvious patterns, formal model validation and goodness-of-fit checks to ensure the linear model is appropriate, rigorous testing of underlying assumptions (like normality and independence of errors), and precise determination of control limit parameters, often through simulation studies.

2.2.2 Phase II: Online Linear Profile Monitoring

Once a stable in-control profile has been reliably established in Phase I, the process transitions to Phase II. Based on Dirbaz, Amiri, Malek, & Salmasnia, (2023) The primary goal of this phase is the real-time detection of process disturbances as quickly as possible.

Control Statistic Calculation and Plotting: In Phase II, new profiles are collected sequentially as the process operates. For each new profile, its parameters are estimated, and a control statistic is calculated based on its deviation from the established Phase I baseline. This statistic is then plotted on the control chart. Signal generation and interpretation is when the plotted point is compared against these control limits.

- If the point falls outside the control limits, a signal is generated. This indicates that the process has likely experienced a significant shift, showing the presence of an assignable cause. This triggers an immediate investigation to identify the root cause and implement corrective actions.
- If the point remains within the control limits, the process is considered to be operating in its in-control state, and monitoring continues with the next acquired profile.

For the calculation of this control statistic, Hotelling's T^2 chart is used, a widely used parameter-based control chart, the specifics of which will be discussed in "Parameter-Based Control Charts" Section of this paper.

Advanced Approaches and Challenges in Phase II Monitoring: Research continually aims to improve the detectability and robustness of Phase II monitoring.

A critical challenge in profile monitoring is the presence of correlation, both within and between profiles. Traditional methods often assume independence, leading to incorrect results. According to Khedmati, & Niaki, (2016) to address between-profile autocorrelation, methods based on statistics like the U statistic have been proposed to eliminate its effect, allowing for the construction of adjusted control charts that monitor the profile parameters effectively. Furthermore, issues such as model misspecification are common, where the assumed parametric model may not perfectly fit the true underlying process. In such cases, parametric approaches can lead to biased estimates, while non-parametric methods may inflate estimated variances. Based on Nassar, & Abdel-Salam, (2022) to overcome these limitations, semi-parametric approaches, such as the Model Robust Regression Technique-2 (MRR2), combine parametric fits with non-parametric residuals fits. These robust methods, often integrated with control charts like the Multivariate CUSUM (MCUSUM) chart, have demonstrated superior performance in detecting shifts across various model misspecification levels, providing more accurate estimations and higher sensitivity.

2.3 Control Charting Techniques for Linear Profiles

Once the in-control profile has been established in Phase I, various control charting techniques can be employed in Phase II to monitor new profiles and detect shifts from the baseline. These methods depend on the detail of the profile being monitored and the primary statistical assumptions.

2.3.1 Parameter-Based Control Charts

Parameter-based control charts represent a fundamental approach to profile monitoring. This method involves fitting a statistical model to each collected profile, estimating its characteristic parameters, and then monitoring these estimated parameters over time. In the parametric approach, the dependent and independent variables are assumed to satisfy a known model (either linear or non-linear), and the charting statistics are then developed based on the estimated model parameters from the data.

Hotelling's T^2 chart

When a linear profile is characterized by multiple parameters such as the intercept and slope, these parameter estimates are often correlated. Monitoring each parameter individually using separate single variable control charts can be insufficient because such charts ignore the correlations among the parameters. A shift in the process might show as a change in the combination of parameters that is not detected by individual charts. Therefore, Hotelling's T^2 chart is a preferred method as it is designed to simultaneously monitor multiple correlated parameters. The T^2 statistic is widely used in multivariate quality control. Williams, Woodall, & Birch (2007) proposed general approaches to the formulation of T^2 statistics based on nonlinear model estimation in Phase I studies.

According to Sabahno & Amiri, (2023)Hotelling's T^2 chart is considered a good option for linear profile monitoring, because it is explicitly developed and compared as one of the core statistical control charts for generalized linear model profiles,. It is further known as the "first and simplest multivariate control chart that can be easily applied in real practices". Its established relevance and effectiveness in the field are highlighted by its consistent use as a benchmark against which new adaptive and machine learning-based charts are compared.

For each new profile i in Phase II, the vector of estimated parameters is denoted as $\hat{\boldsymbol{\beta}}_i$. From Phase I analysis, the in-control vector of parameters is estimated as $\hat{\boldsymbol{\beta}}_0$, and the estimated in-control covariance matrix of these parameters is $\hat{\boldsymbol{\Sigma}}_{\hat{\boldsymbol{\beta}}_0}$. The Hotelling's T^2 statistic for monitoring the i-th profile's parameters is given by:

$$T^2 = (\hat{\boldsymbol{\beta}}_i - \hat{\boldsymbol{\beta}}_0)^T \hat{\boldsymbol{\Sigma}}_{\hat{\boldsymbol{\beta}}_0}^{-1} (\hat{\boldsymbol{\beta}}_i - \hat{\boldsymbol{\beta}}_0) \tag{1}$$

In this equation:

- $\hat{\boldsymbol{\beta}}_i$: This is a vector of the same intercept and slope estimated parameters for the current (new) profile i. For a simple linear profile, this would be $\begin{pmatrix} \hat{\beta}_{0i} \\ \hat{\beta}_{1i} \end{pmatrix}$.

- $\hat{\boldsymbol{\beta}}_0$: This is the known in-control mean vector of parameters, obtained from the stable historical data in Phase I.. For a linear profile, this represents the target intercept and slope $\begin{pmatrix} \hat{\beta}_{00} \\ \hat{\beta}_{10} \end{pmatrix}$. This vector is used as the center point for the T^2 statistic.

- $\hat{\boldsymbol{\Sigma}}^{-1}_{\hat{\boldsymbol{\beta}}_0}$: This is the inverse of the estimated in-control variance - covariance matrix of the parameter estimates denoted as **S** in section '2.2.1'. Its inverse is crucial for weighting the deviations of the current parameters from the target parameters, accounting for their interdependencies.

Under the assumption that the estimated parameters $\hat{\boldsymbol{\beta}}_i$ follow a multivariate normal distribution when the process is in control, the T^2 statistic follows an approximate chi-squared (χ^2) distribution. Specifically, for large samples (large number of observations per profile, n, and a large number of Phase I profiles, m), T^2 eventually behaves liek a χ^2_p where p is the number of parameters being monitored. The upper control limit for the T^2 chart is thus given by:

$$UCL = \chi^2_{\alpha,p}$$

2.3.2 Other Approaches

Single Variant Exponentially Weighted Moving Average (EWMA) Charts

A typical linear profile $Y_{ij} = \beta_0 + \beta_1 X_i + e_{ij}$ has intercept (β_0) and slope (β_1) estimators that are often correlated, particularly when the average of X-values ($\bar{X}$) is not zero. This correlation complicates monitoring because the parameters cannot be assessed independently. While Hotelling's T^2 chart can handle this correlation, Kim et al. wanted to simplify the analysis.

Kim, Mahmoud, and Woodall (2003) propose an alternative Phase II monitoring strategy that simplifies the analysis by "coding" independent variables so that their average becomes zero. In simpler terms, by shifting the X-axis so its center is at zero, they mathematically decouple the intercept and slope estimates, allowing for simpler, individual monitoring. This transformation leads to a new form of the linear model:

$$Y_{ij} = B_0 + B_1 X'_i + e_{ij}$$

Where:

- $B_0 = \beta_0 + \beta_1 \bar{X}$ is the new intercept.
- $B_1 = \beta_1$ is the slope (which remains the same).
- $X'_i = X_i - \bar{X}$ are the centered independent variable values.

In this model, the least squares estimator for B_0 is simply $b_{0j} = \bar{y}_j$, and the estimator for B_1 (b_{1j}) remains the same as β_{1j}. Crucially, with this coding, the estimators b_{0j} and b_{1j} become independent random variables. They are normally distributed with means B_0 and B_1, and variances $\frac{\sigma^2}{n}$ and $\frac{\sigma^2}{S_{xx}}$, respectively. This independence allows for the use of separate univariate control charts for each parameter without concern for correlation effects.

This method uses three separate single variant EWMA charts to detect shifts in the parameters.

a. EWMA Chart for Y-intercept (B_0)

The EWMA statistic for the Y-intercept, $EWMA_I(j)$, is calculated as:

$$EWMA_I(j) = qb_{0j} + (1-q)EWMA_I(j-1)$$

with a smoothing constant q ($0 < q \leq 1$) and initial value $EWMA_I(0) = B_0$.

b. EWMA Chart for Slope (B_1)

The EWMA statistic for the slope, $EWMA_S(j)$, is calculated as:

$$EWMA_S(j) = qb_{1j} + (1-q)EWMA_S(j-1)$$

with smoothing constant q and initial value $EWMA_S(0) = B_1$.

c. EWMA Chart for Error Variance (σ^2)

The EWMA statistic, $EWMA_E(j)$, is calculated as:

$$EWMA_E(j) = \max\{q\ln(MSE_j) + (1-q)EWMA_E(j-1), \ln(\sigma^2)\}$$

with smoothing constant q and initial value $EWMA_E(0) = \ln(\sigma^2)$ (often set to 0 if σ^2 is assumed to be 1 without loss of generality).

The combined use of these three EWMA charts offers easier diagnosis of process changes compared to other methods. For instance, a change in the Y-intercept suggests a bias, a change in the slope could indicate a dilation or contraction of the measurement scale, and an increase in error variance points to greater inaccuracies.

Residual-Based Control Charts

This approach involves analyzing the residuals, which are the differences between the observed dependent values and the values predicted by the fitted profile model. For a linear profile, after fitting the line $Y = \beta_0 + \beta_1 X$, you calculate the residuals $e_{ij} = y_{ij} - \beta_0 - \beta_1 x_i$. If the process is in control, these residuals should ideally exhibit random behavior around zero, indicating that the model captures the process's functional relationship. Deviations from this expected pattern, such as increasing variability in the residuals, can signal a process shift. Control charts can be constructed for various summary statistics of the residuals, such as the Root Mean Squared Error or individual charts on the residuals themselves to monitor their patterns. For example, some control schemes have been proposed based on residuals to monitor simple linear profiles in the presence of between-profile correlations. According to Based on Nassar, & Abdel-Salam, (2022) the residuals are obtained after fitting models and estimating fixed and random effects using a parametric approach, and these residuals can then be further utilized in non-parametric or semi-parametric approaches for fitting and monitoring.

2.4 Performance Measures for Linear Profiles

The most commonly used performance measure in profile monitoring, and specifically highlighted in the context of linear profiles by Kim, Mahmoud, and Woodall (2003), is the Average Run Length (ARL).

- **In-Control ARL ($\mathbf{ARL_0}$):** This represents the average number of samples profiles that are expected to be collected when the process is operating in statistical control before a false alarm occurs. A higher ARL_0 is desirable because it indicates a lower rate of false alarms, reducing unnecessary investigations and process adjustments. For example, studies often aim for an ARL_0 of 200.
- **Out-of-Control ARL ($\mathbf{ARL_1}$):** This represents the average number of samples profiles that are expected to be collected after a true shift has occurred in the process before the control chart signals an out-of-control condition. A lower ARL_1 is highly desirable as it signifies quicker detection of real process changes, allowing for corrective action and minimizing the production of defect products.

By analyzing ARL values across different shift sizes - in the intercept, slope, or error variance of a linear profile, researchers can demonstrate which monitoring methods offer better performance for small, moderate, or large shifts.

3 Practical (R simulations) on Linear Profile Monitoring

This section provides a practical illustration of linear profile monitoring through R simulations. A synthetic dataset will be generated to demonstrate the key steps involved in establishing an in-control profile in Phase I and subsequently monitoring new profiles in Phase II using the Hotelling's T^2 control chart for detecting shifts in profile parameters.

3.1 Description of the Dataset

For this simulation, a synthetic dataset representing linear profiles was generated. This approach was chosen due to its flexibility in controlling process parameters and introducing specific types of shifts, thus providing clear demonstrations of the profile monitoring. While real-world applications would use observed data, simulation allows for a good visual representation of control charts. The data consists of multiple profiles, each characterized by a linear relationship between a dependent variable (y) and a single independent variable (x). The underlying true linear model for each observation y_i at independent variable x_i is given by the following, where $\epsilon_i \sim N(0, \sigma^2)$ represents the random error:

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

```r
# generate y values for phase I profiles
phase1_data$y <- true_beta0_in_control + true_beta1_in_control * phase1_data$x +
  rnorm(m1_phase1 * n, mean = 0, sd = true_sigma_error)
```

Phase I Data (In-Control): A total of $m_1 = 30$ in-control profiles were generated. For each profile, $n = 10$ observations were collected at fixed x values ($x = \{1, 2, \ldots, 10\}$). The true in-control parameters were set as $\beta_0 = 5$ (intercept) and $\beta_1 = 2$ (slope), with an error standard deviation $\sigma = 0.5$. The Type I error rate, α, which represents the probability of a false alarm when the process is in-control, is set to 0.0027. For a T^2 chart, this α value directly determines the UCL based on the chi-squared distribution, aiming for a high in-control ARL ($\text{ARL}_0 \approx 1/\alpha \approx 370.37$).

```r
set.seed(123)
n <- 10        # number of observations per profile
x_values <- 1:n # fixed observation values for all profiles

# true in-control parameters
true_beta0_in_control <- 5
true_beta1_in_control <- 2
true_sigma_error <- 0.5 # standard deviation of the error term

# phase I: data generation and establishment
m1_phase1 <- 30
```

Phase II Data (Monitoring with Shift): A total of $m_2 = 20$ Phase II profiles were generated. The first 10 profiles remained in-control, adhering to the original $\beta_0 = 5$ and $\beta_1 = 2$. For the next 10 profiles, a shift was artificially introduced in the slope parameter (β_1). Specifically, β_1 changed from its in-control value of 2 to an out-of-control value of 2.5. This particular shift was chosen to clearly illustrate the T^2 chart's ability to detect changes in the profile's slope.

```r
# phase II: data generation with a shift
m2_phase2 <- 20 # number of profiles in phase II
shift_start_profile <- 11 # shift starts at this profile number
shifted_beta1 <- 2.5    # new shifted value
```

3.2 Information about the Selected R Packages

The simulation and analysis mainly uses functions available in base R. The `stats` package provides the `lm()` function for fitting linear models, which is central to parameter estimation. Matrix operations required for calculating Hotelling's T^2 statistic like the matrix multiplication, inversion and transpose are also directly supported in base R. The `ggplot2` package was used for creating the control charts.

3.3 Application of Profile Monitoring and Steps

3.3.1 Phase I: Profile Establishment

The goal in Phase I is to define the "normal" operating state of the linear profile.

Parameter Estimation: For each of the $m_1 = 30$ historical profiles, a linear regression model (`lm(y x, data = df)`) was fitted. The `lm()` function in R, by default, utilizes the Ordinary Least Squares method for parameter estimation. This process outputs a set of parameter estimators ($\hat{\beta}_0$ and $\hat{\beta}_1$) for each individual profile. These estimators represent the best-fit intercept and slope for each respective historical profile.

```r
# phase I parameter estimation and control Limit calculation
# fit linear model to each phase I profile
phase1_models <- lapply(split(phase1_data, phase1_data$profile_id),
                        function(df) lm(y ~ x, data = df))

# extract parameter estimates (beta0_hat, beta1_hat) for each phase I profile
beta_hats_phase1 <- t(sapply(phase1_models, coef)) # transpose to get m1 x p
    matrix

# calculatingg the overall in-control parameter vector (mean of Phase I estimates)
# this is the estimate for beta_0 in the T^2 formula
beta_0_hat_overall <- colMeans(beta_hats_phase1)
```

Variance-Covariance Matrix: From this collection of m_1 individual parameter estimates, the overall variance-covariance matrix of estimators was calculated:

```r
# this is our estimate for Sigma_beta_0 in the T^2 formula
Sigma_beta_0_hat <- cov(beta_hats_phase1)
```

Retrospective Analysis: A Hotelling's T^2 chart was applied to the Phase I data to perform retrospective analysis. This involves plotting the T^2 statistic for each historical profile against a calculated UCL. The purpose is to identify any historical profiles that might have been out-of-control, thus potentially biasing the established baseline. For this simulation, all Phase I points are expected to be within limits, confirming a stable baseline. The UCL for Phase I is then determined using the `qchisq` function based on the chosen α (0.0027) and the number of parameters p (2). This ensures the chart is calibrated for a specific false alarm rate during retrospective analysis and subsequent Phase II monitoring.

```r
# determine Hotelling's T^2 control limit for Phase II
# p is the number of parameters being monitored
p <- 2
alpha <- 0.0027 # standard alpha

UCL_T2 <- qchisq(1 - alpha, df = p)
```

3.3.2 Phase II: Online Profile Monitoring

After establishing the stable baseline in Phase I, Phase II involves continuously monitoring new profiles to detect shifts.

```r
# generate y values for phase II profiles, introducing a shift
#new shifted value (shifted_beta1) is 2.5, shift start profile number (shift_start_profile) is 11, number of profiles (m2_phase2)is 20
for (i in 1:m2_phase2) {
  current_beta0 <- true_beta0_in_control
  current_beta1 <- ifelse(i >= shift_start_profile, shifted_beta1, true_beta1_in_control)
  start_row <- (i - 1) * n + 1
  end_row <- i * n
  phase2_data$y[start_row:end_row] <- current_beta0 + current_beta1 * phase2_data$x[start_row:end_row] +
    rnorm(n, mean = 0, sd = true_sigma_error)}
```

Estimation of New Profiles: New profiles are conceptually genrated for Phase II. For each new profile, its parameters - intercept and slope, are estimated using the same linear model fitting process as in Phase I.

Control Statistic Calculation: For each new profile, a control statistic (Hotelling's T^2) is calculated. This calculation uses the new profile's estimated parameters, compared against the in-control parameters (`beta_0_hat_overall`) and their variance-covariance matrix (`Sigma_beta_0_hat`) established in Phase I.

```r
# calculate Hotelling's T^2 for phase II profiles
T2_phase2_values <- numeric(m2_phase2)
for (i in 1:m2_phase2) {
  beta_diff <- matrix(beta_hats_phase2[i, ] - beta_0_hat_overall, ncol = 1)
  T2_phase2_values[i] <- t(beta_diff) %*% solve(Sigma_beta_0_hat) %*% beta_diff}
```

Plotting on Control Chart: The calculated T^2 statistic for each new profile is then plotted on the control chart. This chart includes the predetermined UCL derived from Phase I analysis.

```r
# phase II: plot Hotelling's T^2
plot_df_phase2 <- data.frame(Profile = 1:m2_phase2, T2 = T2_phase2_values)
ggplot(plot_df_phase2, aes(x = Profile, y = T2)) +
  geom_line(color = "blue") +
  geom_point(color = "blue") +
  geom_hline(yintercept = UCL_T2, linetype = "dashed", color = "red", linewidth = 1) +
  geom_vline(xintercept = shift_start_profile - 0.5, linetype = "dotted", color = "darkgreen", linewidth = 1) +
  annotate("text", x = max(plot_df_phase2$Profile) * 0.9, y = UCL_T2 * 1.05,
           label = paste("UCL =", round(UCL_T2, 2)), color = "red", size = 4) +
  annotate("text", x = shift_start_profile - 1, y = max(T2_phase2_values, UCL_T2) * 0.95,
           label = "Shift Starts", color = "darkgreen", hjust = 1, size = 3) +
  labs(title = "Phase II: Hotelling's T  Chart for Linear Profiles (Shift in Slope)",
       y = "Hotelling's T  Statistic", x = "Profile Number") +
  theme_minimal() +
  ylim(0, max(T2_phase2_values, UCL_T2) * 1.1)
```

3.4 Explanation of Results

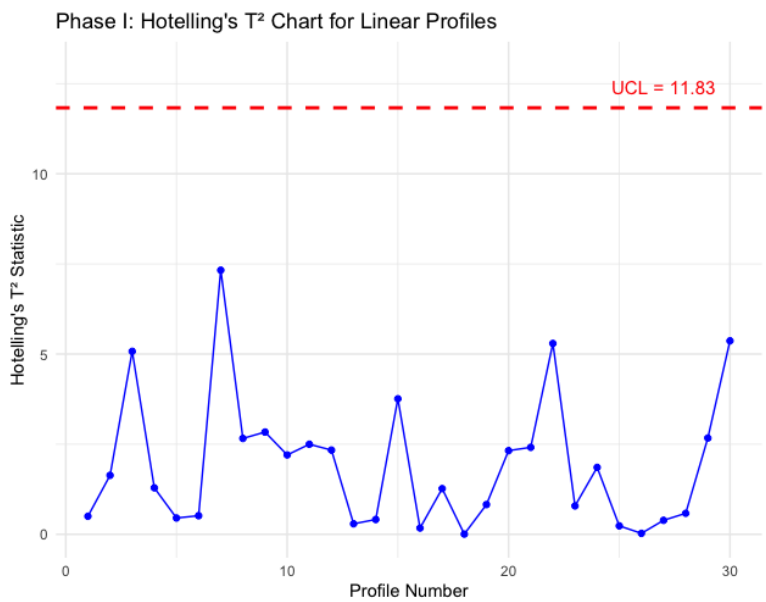

Figure 2: Phase I: Hotelling's T^2 Chart for Linear Profiles

This plot displays the Hotelling's T^2 statistic for each of the 30 simulated Phase I profiles. The blue points represent the calculated T^2 values for each profile. The red dashed line indicates the UCL at 11.83. All points on this chart fall below the UCL, confirming that the historical data used to establish the baseline was indeed in statistical control. This retrospective analysis validates the selected in-control state of the process.

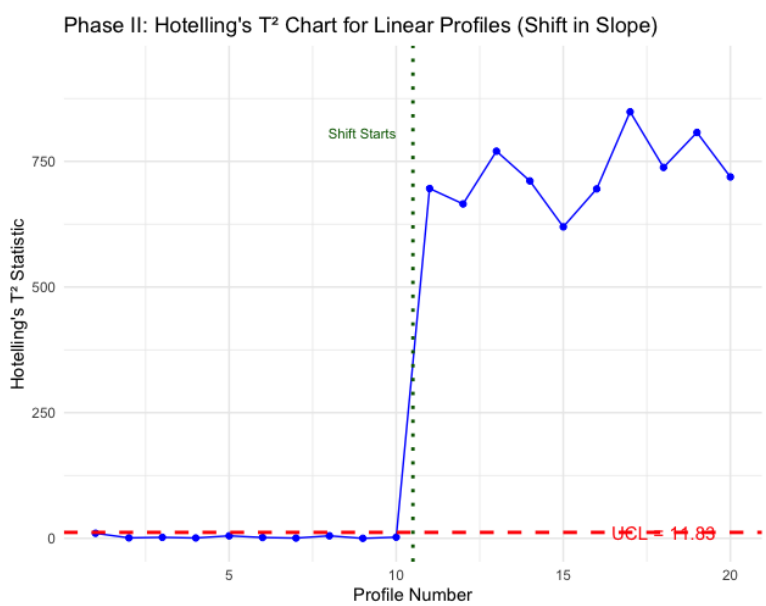

Figure 3: Phase II: Hotelling's T^2 Chart for Linear Profiles

This plot displays the Hotelling's T^2 statistic for the 20 sequentially collected Phase II profiles, plotted against its UCL at 11.83. The vertical dotted green line, labeled "Shift Starts", clearly marks profile number 11 as the point where the slope parameter was artificially increased.

- **Pre-Shift Behavior (Profiles 1-10):** For the first 10 profiles, which were generated under in-control conditions, the Hotelling's T^2 statistic values remain very low, fluctuating close to zero and well below the UCL. This indicates that the process was stable and in control before the shift.
- **Post-Shift Behavior (Profiles 11-20):** Immediately after the shift in the slope was introduced at profile 11, the T^2 statistic values exhibit a dramatic increase. The point at Profile Number 11 itself shows a T^2 value significantly above the UCL, generating a clear out-of-control signal. The code geneates [`Out-of-control signals detected at profiles:  11 12 13 14 15 16 17 18 19 20`]

These results are good because the chart clearly and promptly detected the introduced shift. The significant increase in theT^2 statistic immediately following the shift shows the chart's sensitivity and ability to correctly identify when the linear profile's characteristics deviate from the established in-control baseline. This quick response of the chart to the actual process change, observed in this specific simulation, illustrates a low run length, this instance validates the utility of Hotelling's $T^2$for monitoring linear profiles.

4 Conclusion

In conclusion, this paper mainly addresses the necessity of profile monitoring as an SPC technique to manage the quality of modern processes where product characteristics are defined by functional relationships rather than single measurements.

This paper has a theoretical overview of linear profile monitoring, detailing its two distinct phases: Phase I, dedicated to establishing a stable in-control baseline through parameter estimation and retrospective analysis, and Phase II, focused on the real-time detection of process disturbances. The practical application through R simulations demonstrated the effectiveness of the Hotelling's T^2 control chart in successfully identifying shifts in linear profile parameters. This method's quick detection of problems means it's important for keeping product quality steady and fixing issues fast in product manufacturing.

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